

Shark Colony Experiment

Shared-ocean ecology (GA + RL) — 2026-06-12T04:08:08

Summary

Summary	
Cycles requested / completed	100 / 100
Founding pod	30
Carrying capacity	100
Peak population	100
Final population	90
Colony went extinct?	no
Best fitness (any cycle)	112.400
Final mean fitness	-7.187
Runtime	4.0 s

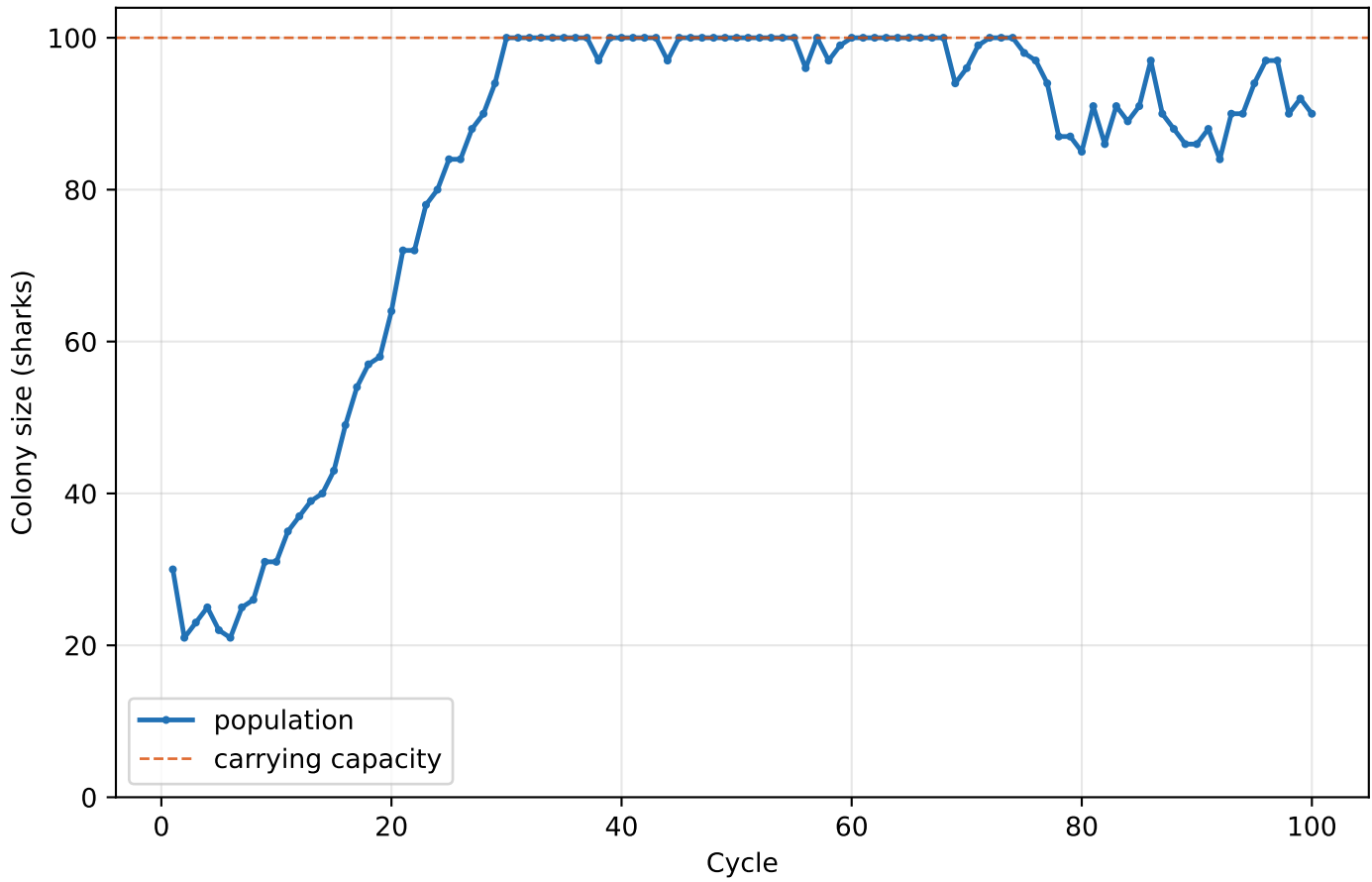
Ecology & competition

Ecology setting	Value
Shared, depleting ocean	True
Fish per shark / pool cap	1.8 / 30
Must eat to breed	True
Fitness-weighted breeding	True
Cull weakest on overcrowding	True
Gestation divisor / brood	6.0 / 2
Steps per life / brain warmup	80 / 400
Mutation rate / seed	0.2 / 0
Run timestamp	2026-06-12T04:08:08

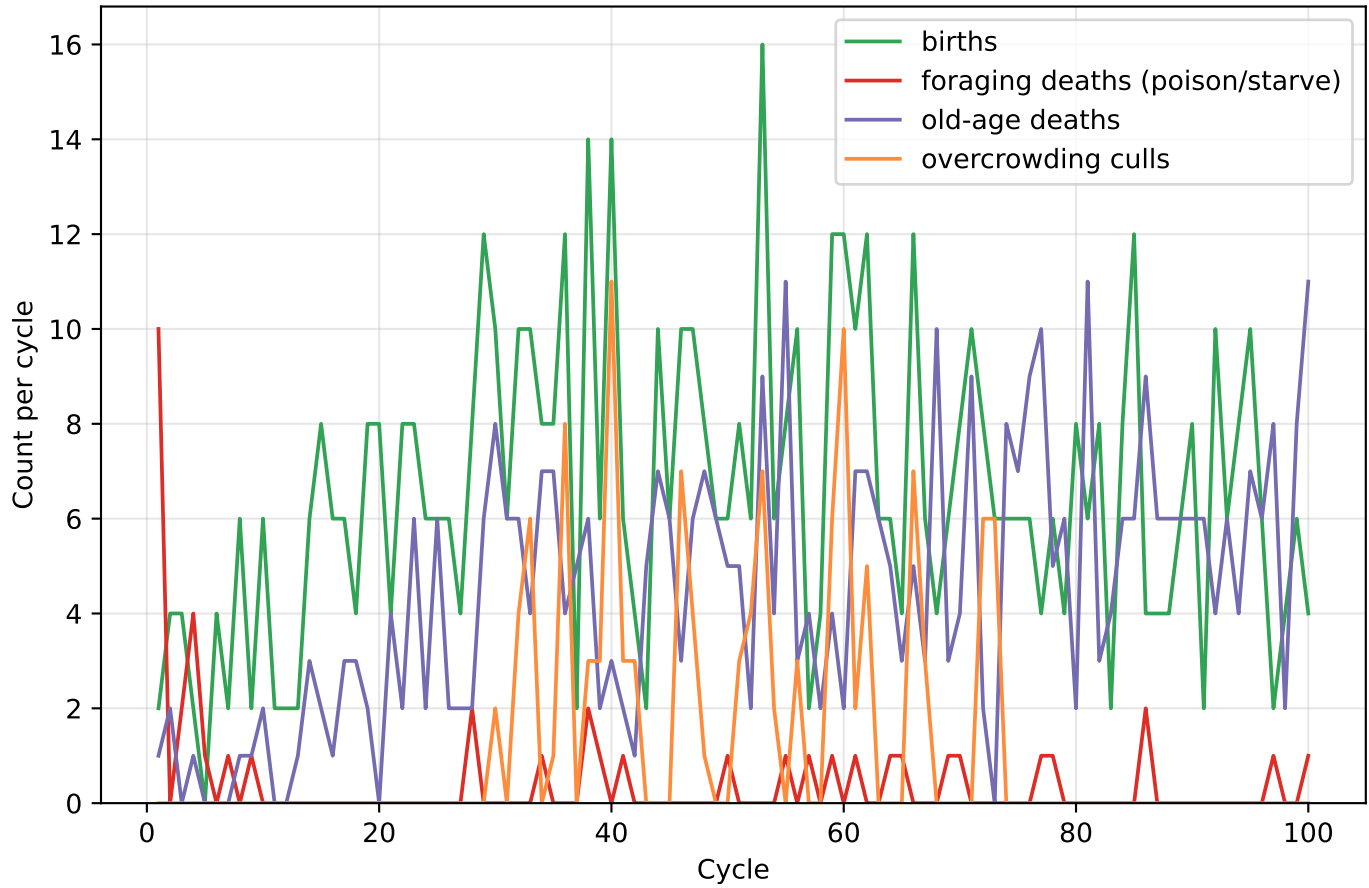
Shared brain (Q-learning)

Shared brain (Q-learning)	Value
States learned	236
Final epsilon (explore rate)	0.061
Epsilon min / decay	0.050 / 0.9950
Alpha (learning rate)	0.100
Gamma (discount)	0.950

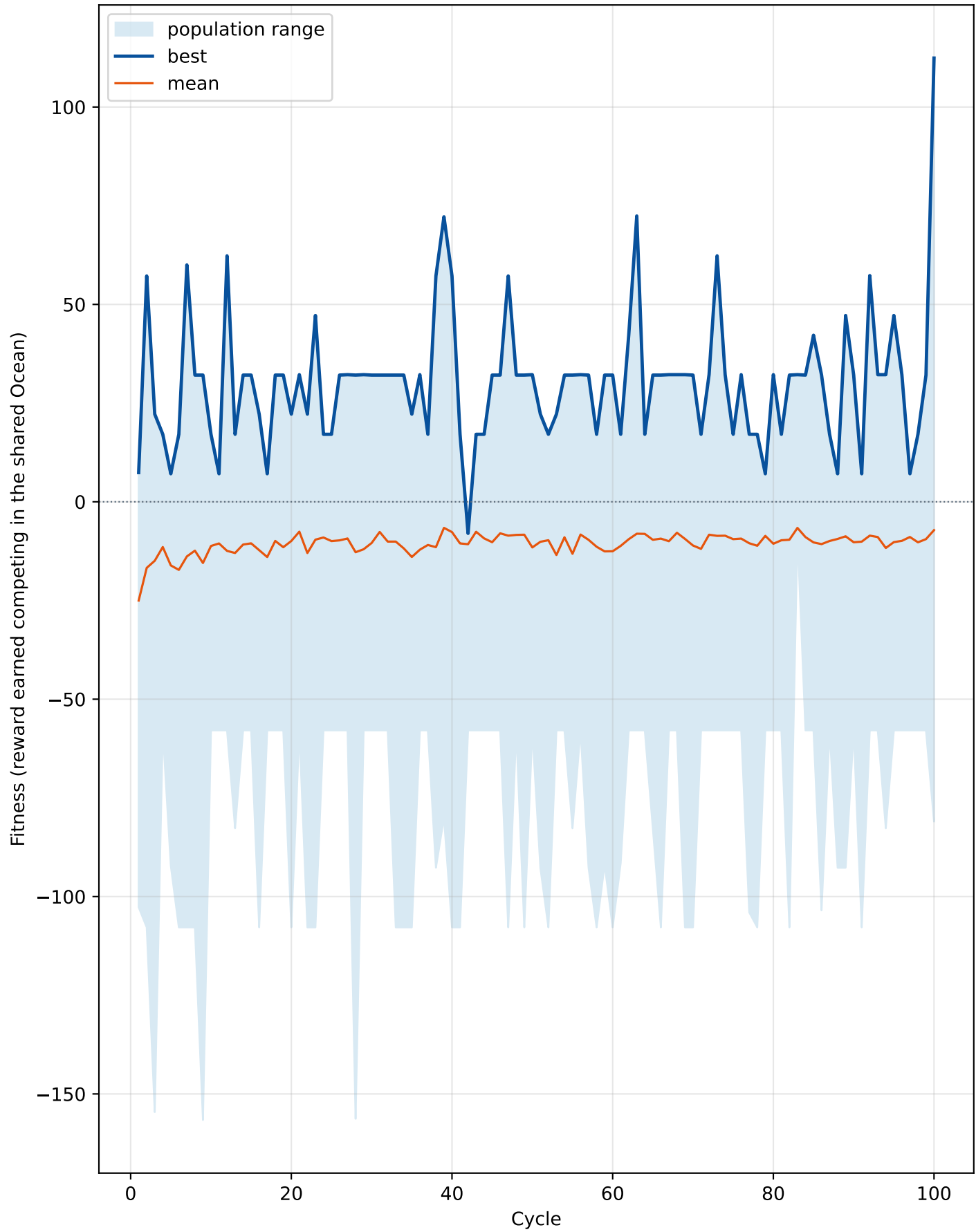
Population over cycles



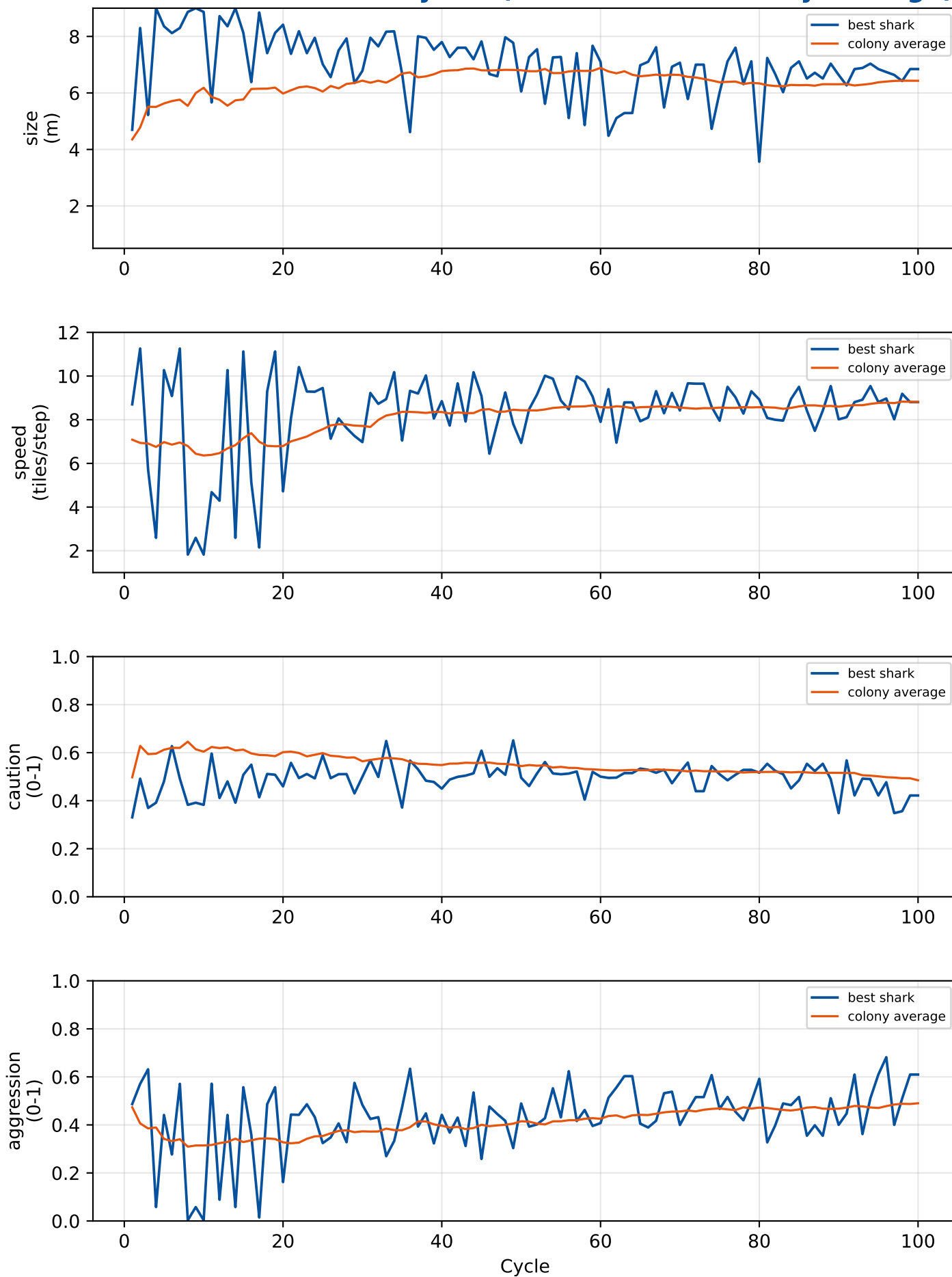
Births & deaths per cycle



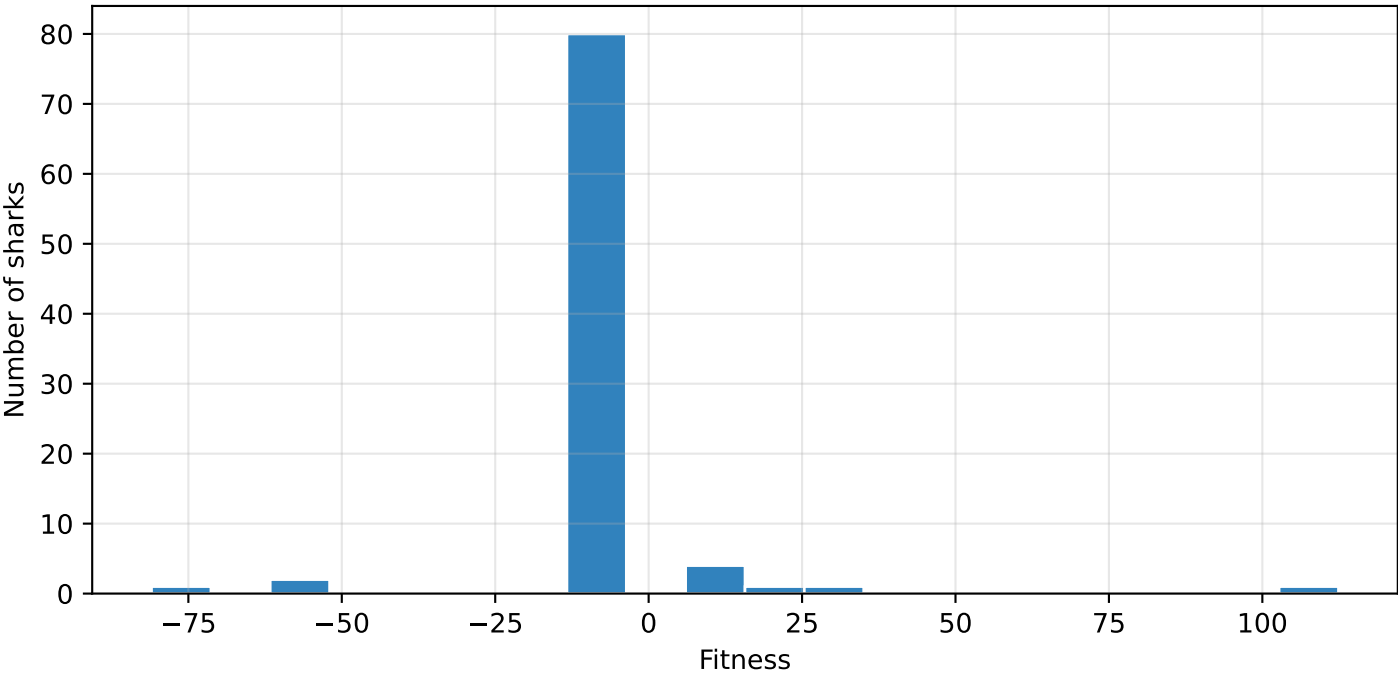
Fitness under competition over cycles



Evolvable traits over cycles (best shark vs. colony average)



Final-cycle fitness distribution (cycle 100)



Best genome found (highest single-cycle reward under competition)

Trait	Value	Unit	Evolved?
size	6.845	m	yes
speed	8.807	tiles/step	yes
caution	0.422	0-1	yes
gestation_period	12.000	months	no (fixed)
aggression	0.610	0-1	yes
field_of_perception	5.000	tiles (radius)	no (fixed)
color_hue	205.000	degrees (HSV hue)	no (fixed)

Per-cycle summary (showing 26 of 100 cycles, every 4)

Cycle	Pop	Size	Speed	Caution	Aggression	Best	Mean	Born	Died	Cull
1	30	4.70	8.70	0.33	0.49	7.4	-25.0	2	11	0
5	22	8.36	10.27	0.48	0.44	7.1	-16.1	0	1	0
9	31	9.00	2.59	0.39	0.06	32.1	-15.5	2	2	0
13	39	8.36	10.27	0.48	0.44	17.1	-13.0	2	1	0
17	54	8.85	2.15	0.41	0.01	7.1	-14.0	6	3	0
21	72	7.38	8.07	0.56	0.44	32.2	-7.6	4	4	0
25	84	7.02	9.45	0.59	0.32	17.1	-10.0	6	6	0
29	94	6.35	7.26	0.43	0.58	32.2	-12.0	12	6	0
33	100	8.17	8.94	0.65	0.27	32.1	-10.1	10	4	6
37	100	8.00	9.20	0.53	0.39	17.1	-10.9	2	5	0
41	100	7.27	7.73	0.49	0.37	17.1	-10.5	6	3	3
45	100	7.82	9.08	0.61	0.26	32.1	-10.2	6	6	0
49	100	7.77	7.81	0.65	0.30	32.1	-8.3	6	6	0
53	100	5.62	10.02	0.56	0.43	22.2	-13.5	16	9	7
57	100	7.41	9.99	0.52	0.42	32.1	-9.6	2	5	0
61	100	4.48	9.40	0.49	0.51	17.1	-11.2	10	8	2
65	100	6.98	7.93	0.53	0.41	32.1	-9.6	4	4	0
69	94	6.93	9.23	0.47	0.54	32.2	-9.4	6	4	0
73	100	7.00	9.65	0.44	0.52	62.3	-8.6	6	0	6
77	94	7.60	9.03	0.51	0.45	17.1	-10.5	4	11	0
81	91	7.24	8.08	0.55	0.33	17.1	-9.7	6	11	0
85	91	7.12	9.51	0.48	0.52	42.2	-10.3	12	6	0
89	86	7.04	9.54	0.49	0.51	47.2	-8.7	6	6	0
93	90	6.88	8.92	0.49	0.36	32.2	-8.9	6	6	0
97	97	6.64	8.02	0.35	0.40	7.1	-8.9	2	9	0
100	90	6.84	8.81	0.42	0.61	112.4	-7.2	4	12	0